CECS/CCCS 111 – Introduction to Programming
University of Jeddah

Course Project: Library management system

Points: 10 | CLO3.1/SO.7

Submission Deadline: 05 Dec 2024

Project Requirements

Build an application for a library management system. The application should include the following features:

1. **Menu Options:**
   Provide a menu that lets a user choose one of the following options:

```
Choose one of the following options:
1 - to add a new member
2 - to add a new book
3 - to print all members' information
4 - to print all books' information
5 - to quit
```

2. **Option 1:**
   If the user selects option 1, the program should prompt the user to enter new member information. Member information should be stored using a `struct` data structure with the following fields:

   - **Member ID** (unique number)
   - **Name**
   - **Phone Number**

3. **Option 2:**
   If the user selects option 2, the program should prompt the user to enter new book information. Book information should be stored using a `struct` data structure with the following fields:

   - **Book ID** (unique identifier)
   - **Title**
   - **Author**
   - **Year of Publication**
   - **Member ID** (representing the member who has currently borrowed the book, or `0` if available)

4. **Option 3:**
   If the user selects option 3, the program should print all members' information in a tabular format, including:

- **Member ID**
- **Name**
- **Phone Number**

- **Option 4:**
  If the user selects option 4, the program should print all books' information in a tabular **Book ID**
- **Title** + **Author** + **Year of Publication**
- **Member ID** (indicating which member has borrowed the book, or display "Available" if no one has borrowed it)

Please note that the program will display the same menu again after performing the required operations in options 1,2,3 and 4.

5. **Option 5:**
   If the user selects option 5, the program should terminate.

Additional Notes:

- Use `struct` to store member and book records.
- Ensure input validation (e.g., unique IDs, non-empty fields).
- Maintain a modular structure using functions for each task (e.g., adding a member, adding a book, printing records).
- Format the output for readability, especially when displaying tables.

Submission Instructions

Step 1: Team Formation

Form your teams of 2-3 students for the course project.
You have to submit your team members' names and student numbers to your Lab Instructor.

Step 2: Project Submission Instructions

To submit your project, one team member should upload a project report containing the following things through the assignment created on BlackBoard:

1. Project report in MS Word (.DOCX file) format that includes the following
   a. Project title
   b. Names and Student IDs of team members
   c. A breakup of tasks across team members (who did what?)
2. Source code (.C file) of your program.

Step 3: Presentation

Ensure that you coordinate with your lab instructor to schedule your project presentation. Each student should do the presentation. Failure to do so will result in a score of 0 for the project.